

SYSTEMATIC_DEBUGGING_2026-04-19

Systematic Debugging Session — 2026-04-19

Revision: 1 **Last modified:** 2026-06-06T00:00:00Z

Applied the `superpowers:systematic-debugging` skill against the 11-issue set from the prior subagent's work. Separated “claimed fixed” from “verified fixed” and found one genuine residual bug beyond the original 11.

Phase 1 — Root cause investigation (evidence gathered, no fixes yet)

Live-stack code checksums match committed `main`; the subagent's changes are live. Evidence per issue:

#	Issue	Verdict	Evidence
1	White frame around dashboard	FIXED	Global SCSS in served CSS: <code>html,body{margin:0;padding:0;background:#0f1a30}</code>
2	Auth indicator doesn't refresh after login	FIXED	Bundle ships <code>openQbitLogin()</code> which passes <code>loadAuthStatus</code> as the success callback
3	Lost per-tracker status chips	FIXED	Bundle contains <code>tracker-chip (×2) + Re-login (×2)</code> ; <code>/api/v1/auth/status</code> returns 4 trackers

#	Issue	Verdict	Evidence
4	WebUI Bridge link Page Unavailable	FIXED	GET /api/v1/bridge/health → {"healthy":true,"port":7188}; bundle has bridge-down-hint
5	qBittorrent WebUI link Unauthorized	FIXED	GET /api/v1/config → qbittorrent_url: "http://localhost:7186" (proxy, not :7185 internal)
6	Search not real-time	FIXED	POST /api/v1/search returns in ~100ms with status:"running", results:[]; SSE /stream/{id} emits event: result_found within 500ms
7	Sources column broken	FIXED	Bundle has sourceStats (×3) + merged-indicator (×4); backend returns sources: [{tracker, seeds, leechers}]
8	Low seeds/leechers	NOT A BUG	Real tracker data; dedup sums correctly. Live search returned seeds=326, leechers=69 from rutracker
9	qBit chip missing username + logout	FIXED	Bundle has qBit Connected, qbit-chip; /auth/status returns username:"admin"; bundle has Re-login
10	Only 2 trackers in results	NOT A BUG	trackers_searched = 40; live logs show per-tracker counts (iptorrents: 49 results, kinozal: 0 results, academictorrents: 0 results); zero matches for a specific query is tracker-side behaviour
11	qBit download appears successful but no torrent added	FIXED — end-to-end verified	Before: 0 torrents. After POST /api/v1/download with Ubuntu magnet: 1 torrent, name=ubuntu-22.04.3-desktop-amd64.iso, hash=dd8255ec..., state=checkingResumeData. Test torrent cleaned up afterwards.

Each “FIXED” verdict came from an independent live probe, not from trusting the subagent’s summary.

Unexpected finding — real bug uncovered by new property test

After the 11-issue verification, running the newly-added property suite surfaced:

```
FAILED tests/property/  
test_deduplicator_properties.py::test_merge_is_order_invariant
```

Hypothesis minimised a counterexample:

```
[(name='0', hash=...0000),  
 (name='0', hash=...0001),  
 (name='00', hash=...0000)]
```

- Forward: produces **one** group of 3 (all merged)
- Reversed: produces **two** groups ({name='00', hash=...0000}, {name='0', hash=...0001})

Different groupings → same results would render differently in the dashboard depending on which tracker's response arrived first.

Phase 2 — Pattern analysis

deduplicator.py::merge_results (line 65) sorted by seed count only:

```
unmatched.sort(key=lambda r: r.seeds, reverse=True)
```

Python's sort is stable. Equal-seed items therefore preserved their input order, and — in the counterexample — all three items had seeds=0. Different input orders produced different sort orders, each of which seeded different merge groups.

Phase 3 — Hypothesis

Adding a deterministic secondary sort key on (link, name) would force equal-seed items to resolve the same way regardless of input order, eliminating the nondeterminism without changing the seed-priority semantics.

Phase 4 — Fix

One-line change:

```
unmatched.sort(key=lambda r: (-r.seeds, r.link or "", r.name or
    ""))
```

Verified: - test_merge_is_order_invariant flips from fail → pass. - All 686 deterministic tests still green. - No other property regressed.

Deployed by restarting qbittorrent-proxy; /health responded 200 after ~3 s.

Total score

- 11 user-reported issues: 9 fixed, 2 “not a bug” (confirmed with evidence).
- 1 additional real bug found by the new property suite, fixed same session.
- Tests: 686 deterministic Python + 182 frontend = **868 passing / 0 failures / 0 skips / 0 warnings**.
- Live stack: healthy, latest code serving.